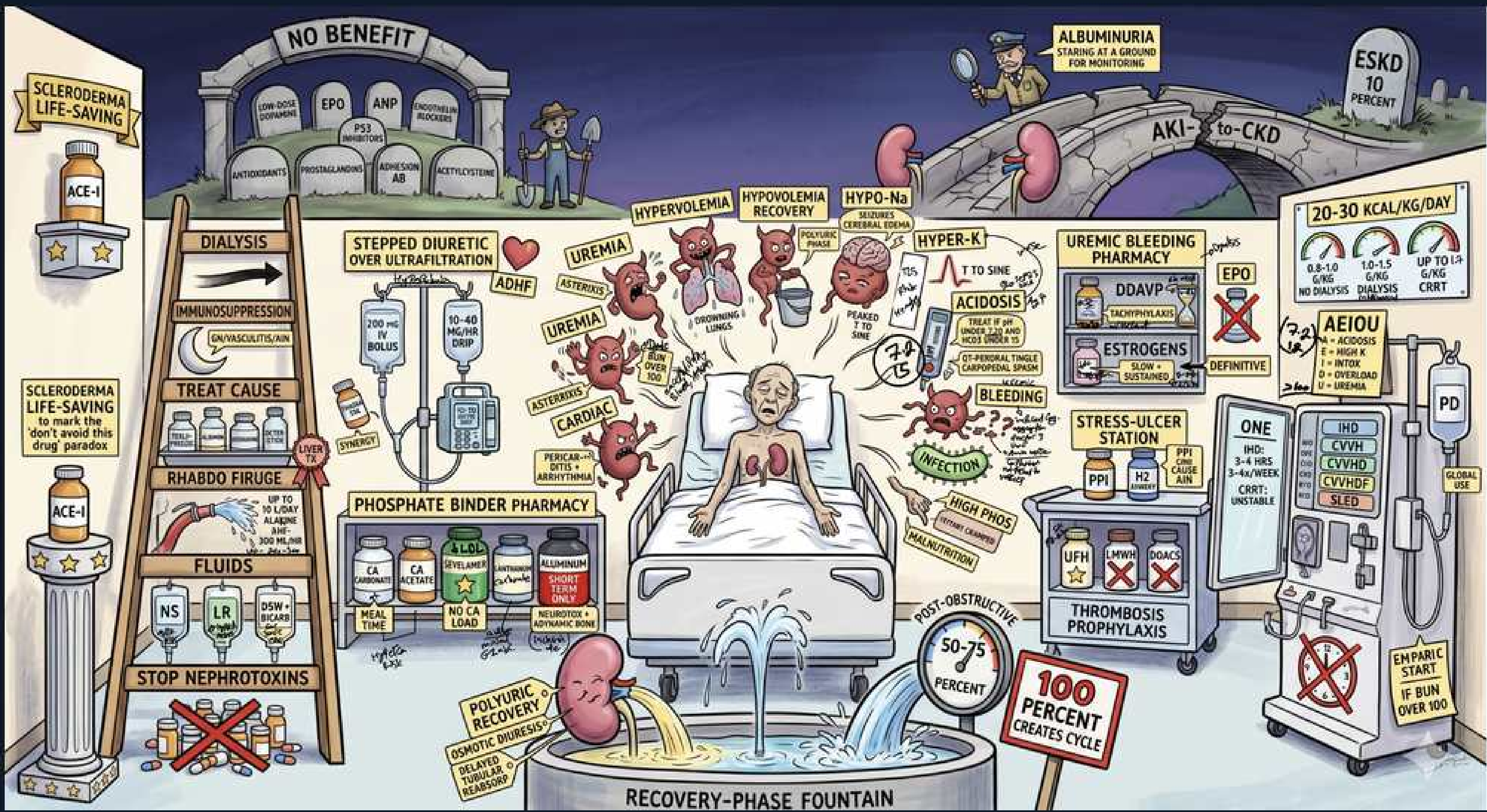
AKI — Complications & Treatment



1. Dialysis indications

WHAT YOU SEE

A ladder whose rungs you climb (treat cause, fluids, stop nephrotoxins) up to DIALYSIS at the top — dialysis is the last rung, reached only after the lower steps fail, not a number on a lab.

WHAT IT ENCODES

Renal replacement therapy (intermittent hemodialysis or CRRT in the unstable ICU patient) is the rescue for AKI complications that are refractory to medical management. There is no proven mortality benefit to 'early/prophylactic' dialysis based on BUN/creatinine numbers alone — start it for the AEIOU clinical indications, not for an arbitrary lab threshold.

BOARD TRAP

Initiating dialysis purely because the creatinine or BUN crosses a number is the wrong answer — recent trials show no benefit to early dialysis without a clinical indication. The discriminator is presence of a refractory AEIOU complication.

2. AEIOU

WHAT YOU SEE

AEIOU box, ESKD 10 percent gravestone

WHAT IT ENCODES

Mnemonic AEIOU for urgent dialysis indications: Acidosis (severe metabolic, pH <7.1 refractory to bicarbonate), Electrolytes (refractory hyperkalemia), Intoxications (dialyzable toxins — methanol, ethylene glycol, salicylates, lithium; mnemonic 'I STUMBLED'), Overload (volume overload/pulmonary edema unresponsive to diuretics), and Uremia (pericarditis, encephalopathy, bleeding). About 10% of severe AKI progresses toward ESKD/CKD.

BOARD TRAP

Forgetting that the dialysis trigger is REFRACTORY complications — e.g., hyperkalemia that responds to medical therapy does NOT need dialysis. The discriminator is failure of, or contraindication to, medical management.

3. Hyperkalemia

WHAT YOU SEE

An angry HYPER-K figure beside a peaked-T-wave ECG tracing — the furious figure and spiking T waves are the most immediately lethal AKI complication.

WHAT IT ENCODES

Hyperkalemia is the most immediately life-threatening AKI complication. ECG progresses: peaked T waves → flattened P/prolonged PR → widened QRS → sine wave → VF/asystole. Treatment order: (1) IV calcium gluconate to stabilize the cardiac membrane (does NOT lower K), (2) shift K intracellularly with insulin+glucose, albuterol, and bicarbonate if acidotic, then (3) REMOVE K with loop diuretics, potassium binders (patiromer, sodium zirconium cyclosilicate), or dialysis.

BOARD TRAP

Two traps: (1) giving insulin/glucose or kayexalate FIRST in a patient with ECG changes — calcium must come first to protect the heart; (2) thinking calcium lowers potassium — it only stabilizes the myocardium and buys time while you shift/remove K.

4. Metabolic acidosis

WHAT YOU SEE

A slumped ACIDOSIS figure labelled with a downward pH — the figure being eaten away depicts the retained acids of a high-anion-gap metabolic acidosis.

WHAT IT ENCODES

AKI causes a high anion-gap metabolic acidosis from retained organic acids, sulfates, and phosphates that the failing kidney cannot excrete. Treat with IV sodium bicarbonate when severe (e.g., pH <7.1–7.2); persistent severe acidosis refractory to bicarbonate is an AEIOU dialysis indication.

BOARD TRAP

Aggressively giving bicarbonate for mild AKI acidosis is wrong — it risks volume overload, hypocalcemia, and hypokalemia. Reserve bicarbonate for severe acidosis; the discriminator for dialysis is REFRACTORINESS, not the mere presence of acidosis.

5. Volume overload

WHAT YOU SEE

HYPERVOLEMIA figure, stepped diuretic / ultrafiltration

WHAT IT ENCODES

Oliguric AKI causes volume overload and pulmonary edema. First-line is high-dose loop diuretics (e.g., furosemide, often a continuous infusion or a stepped/escalating regimen); if the patient is truly diuretic-resistant, use ultrafiltration or dialysis to remove fluid.

BOARD TRAP

Giving IV fluids to an AKI patient regardless of volume status is WRONG — fluids help only PRERENAL/hypovolemic AKI. In a volume-overloaded oliguric patient, fluids worsen pulmonary edema. The discriminator is assessing actual volume status before prescribing fluids vs diuresis.

6. Uremia

WHAT YOU SEE

A toxin-laden UREMIA figure next to the ADHF/overload scene — the poisoned figure embodies accumulated nitrogenous waste causing encephalopathy, pericarditis, and platelet dysfunction.

WHAT IT ENCODES

Uremia from accumulated nitrogenous waste causes encephalopathy (asterixis, confusion, seizures), pericarditis (with a friction rub and risk of tamponade), platelet dysfunction with bleeding, nausea, and pruritus. Uremic pericarditis and encephalopathy are firm indications for urgent dialysis.

BOARD TRAP

Treating uremic pericarditis with NSAIDs/colchicine as for viral pericarditis is wrong — the definitive treatment is DIALYSIS (and NSAIDs are nephrotoxic). The discriminator is the uremic etiology, which calls for renal replacement, not anti-inflammatories.

7. Uremic bleeding

WHAT YOU SEE

UREMIC BLEEDING PHARMACY, DDAVP, estrogens

WHAT IT ENCODES

Uremic toxins impair platelet adhesion/aggregation (qualitative platelet dysfunction) causing mucosal/GI bleeding and a prolonged bleeding time despite a normal platelet count and normal PT/PTT. Treatment ladder: DDAVP (desmopressin, rapid onset, releases vWF), cryoprecipitate, conjugated estrogens (slower but longer-lasting), correct anemia (raise Hct to ~30%), and dialysis to remove uremic toxins.

BOARD TRAP

Transfusing platelets for uremic bleeding is the wrong answer — the platelet COUNT is normal and the defect is functional, so transfused platelets quickly become dysfunctional too. The discriminators/fixes are DDAVP, dialysis, and correcting anemia, not platelet transfusion.

8. Treat the cause

WHAT YOU SEE

TREAT CAUSE rung, immunosuppression

WHAT IT ENCODES

The core of AKI management is identifying and reversing the underlying cause: relieve obstruction (postrenal), restore perfusion (prerenal), treat sepsis, and give immunosuppression for immune-mediated intrinsic disease (e.g., steroids for AIN or rapidly progressive GN). Everything else is supportive.

BOARD TRAP

Jumping to dialysis or generic supportive care without working through the prerenal/intrinsic/postrenal framework misses reversible causes — e.g., a bladder-outlet obstruction fixed by a Foley catheter. Always exclude obstruction with a bladder scan/ultrasound.

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9. Give fluids

WHAT YOU SEE

FLUIDS rung, NS / LR / RA bags

WHAT IT ENCODES

Restore renal perfusion with isotonic crystalloid (normal saline or a balanced solution like lactated Ringer's) in prerenal/hypovolemic AKI. Balanced crystalloids may cause less hyperchloremic acidosis and AKI than large-volume saline. Fluids are the treatment for prerenal AKI specifically.

BOARD TRAP

Believing fluids help ALL AKI is wrong — in volume-overloaded or oliguric intrinsic AKI, fluids cause pulmonary edema. Fluids are for the HYPOVOLEMIC/prerenal patient; the discriminator is documented volume depletion.

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10. Stop nephrotoxins

WHAT YOU SEE

STOP NEPHROTOXINS rung, red X on bottles

WHAT IT ENCODES

Stop and avoid nephrotoxins: NSAIDs, iodinated contrast, aminoglycosides, amphotericin B, ACE-I/ARBs (in most AKI), and renally cleared drugs that need dose adjustment. Review the full med list and renally dose-adjust everything in AKI to prevent further injury and drug toxicity.

BOARD TRAP

Continuing an ACE-I/ARB or NSAID in AKI 'because the patient was on it chronically' is wrong — these worsen AKI in most settings. The big EXCEPTION is scleroderma renal crisis, where ACE-I is life-saving and must NOT be stopped.

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11. ACE-I scleroderma

WHAT YOU SEE

SCLERODERMA LIFE-SAVING ACE-I, 'don't avoid this drug'

WHAT IT ENCODES

Scleroderma renal crisis presents with malignant hypertension, AKI, and MAHA (schistocytes) in a patient with diffuse systemic sclerosis. ACE inhibitors (captopril, titrated rapidly) are LIFE-SAVING and the treatment of choice — this is the major exception to the rule of avoiding ACE-I in AKI. Risk is increased by high-dose corticosteroids.

BOARD TRAP

Avoiding or stopping ACE-I in scleroderma renal crisis (the usual AKI reflex) is the deadly wrong answer — here ACE-I dramatically improves survival. Also do NOT give high-dose steroids in scleroderma, as they precipitate the crisis.

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12. No benefit

WHAT YOU SEE

NO BENEFIT gravestones: EPO, ANP, dopamine

WHAT IT ENCODES

Therapies with NO proven benefit (and potential harm) in AKI: low-dose 'renal-dose' dopamine, fenoldopam, ANP/natriuretic peptides, mannitol, and loop diuretics given FOR renal protection or recovery. Diuretics may help fluid management but do not improve renal recovery or mortality and don't convert AKI to a milder course in outcome terms.

BOARD TRAP

Choosing low-dose dopamine 'to protect the kidney' or improve urine output is a classic wrong answer — it does not preserve renal function or reduce mortality and risks arrhythmia. None of these agents change AKI outcomes.

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13. Stress-ulcer prophylaxis

WHAT YOU SEE

STRESS-ULCER STATION, PPI, thrombosis prophylaxis

WHAT IT ENCODES

Supportive ICU care for critically ill AKI patients includes stress-ulcer prophylaxis (PPI or H2 blocker for those with risk factors such as mechanical ventilation >48h or coagulopathy) and VTE/thrombosis prophylaxis. Avoid unnecessary prophylaxis in low-risk patients.

BOARD TRAP

Giving stress-ulcer prophylaxis to every floor patient is over-treatment and may raise C. difficile/pneumonia risk — reserve it for ICU patients with genuine risk factors. Note PPIs are themselves a leading cause of AIN, so use judiciously.

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14. Recovery diuresis

WHAT YOU SEE

RECOVERY-PHASE FOUNTAIN, polyuria, 50-75 percent

WHAT IT ENCODES

In the recovery phase of ATN and after relief of obstruction, a POLYURIC (post-obstructive) diuresis occurs as tubular function returns but concentrating ability lags. Watch for and replace fluid and electrolyte (K, Mg, phosphate) losses; most patients (~50–75% or more) recover baseline renal function, with creatinine cycling back toward normal.

BOARD TRAP

Reflexively matching urine output milliliter-for-milliliter during post-obstructive diuresis can perpetuate it and cause volume depletion — replace a fraction (e.g., ~50–75% of losses) and monitor volume status. Failing to replace electrolytes risks dangerous hypokalemia/hypomagnesemia.

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15. Phosphate binder pharmacy

WHAT YOU SEE

PHOSPHATE BINDER PHARMACY shelf, calcium/sevelamer bottles

WHAT IT ENCODES

Hyperphosphatemia from reduced renal excretion (worsened by tumor lysis or rhabdomyolysis) is managed with dietary phosphate restriction and oral phosphate binders taken WITH meals — non-calcium binders (sevelamer, lanthanum) or calcium-based binders (calcium acetate/carbonate). Severe hyperphosphatemia drives hypocalcemia and may require dialysis.

BOARD TRAP

Giving IV calcium for the hypocalcemia of hyperphosphatemia (e.g., in tumor lysis) can precipitate calcium-phosphate in tissues and worsen AKI — treat the phosphate first. Also, binders only work if taken WITH food; given between meals they don't bind dietary phosphate.

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16. Albuminuria monitoring

WHAT YOU SEE

ALBUMINURIA staging gauge / 'staging + AKI-to-CKD monitoring' bridge

WHAT IT ENCODES

AKI is a major risk factor for progression to CKD (the 'AKI-to-CKD' transition, ~10% reach ESKD). Survivors need follow-up with serial creatinine/eGFR and albuminuria (urine albumin-to-creatinine ratio) staging, because persistent albuminuria and incomplete recovery predict progression. Nephrology follow-up after significant AKI is recommended.

BOARD TRAP

Discharging a recovered-AKI patient with no renal follow-up because the creatinine 'came back to normal' is wrong — apparent recovery can mask reduced renal reserve and ongoing albuminuria. The discriminator is monitoring eGFR and ACR over time, not a single normal creatinine.

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17. Nutrition 20-30 kcal/kg

WHAT YOU SEE

20-30 KCAL/KG/DAY nutrition board, protein/calorie targets

WHAT IT ENCODES

Nutritional support in AKI targets roughly 20–30 kcal/kg/day with adequate protein (≈0.8–1.0 g/kg/day non-catabolic, up to 1.5–1.7 g/kg/day in critically ill or those on CRRT, which removes amino acids). Enteral nutrition is preferred; avoid severe protein restriction, which worsens catabolism without preventing dialysis.

BOARD TRAP

Severely restricting protein to 'rest the kidney' or delay dialysis is the wrong answer — it promotes malnutrition and catabolism. Adequate calories and protein (more, not less, on CRRT) are the goal; dialysis is driven by AEIOU indications, not by diet.
